HUD — ESP32-S3 Super Mini + NV3007 2.79″

428 × 142 · Firmware GPIO mapping · Display via 5 → 3.3 V DC-DC · TJA1051T/3 CAN module

Based on gateway_supermini.svg · Receive-only CAN · Phone BLE settings and OTA

Vehicle

ACC +12 V (terminal 15)

GND (terminal 31)

CAN-H
MLB-Evo: violet
MQB: orange-green

CAN-L
MLB-Evo: orange-brown
MQB: orange-brown

MLB-Evo: Infotainment CAN
MQB: Comfort CAN
500 kbit/s

CAN-H / CAN-L:
Twisted pair

Module 120 Ω:
Disable / remove

F1

0.5–1 A

DC-DC 12 → 5 V

Powered from ACC

GND

470 µF + 100 nF

5V

Display power

DC-DC input from the shared 5V rail

5V → DC-DC 5 → 3.3 V → 3V3_D

47 µF + 100 nF

ESP32-S3 Super Mini

4 MB Flash / 2 MB PSRAM

5V	5V
GND	GND
SPK	GPIO7 · SPK
FL_CS	GPIO10 · FL CS
FL_MOSI	GPIO11 · FL MOSI
FL_SCK	GPIO12 · FL SCK
FL_MISO	GPIO13 · FL MISO
LDR	GPIO14 · LDR
	GPIO9 · CAN TX
	GPIO8 · CAN RX
Board 3V3	3V3_P
GPIO1 · SCK	
GPIO2 · MOSI	
GPIO5 · CS	
GPIO4 · DC	
GPIO3 · RST	
GPIO6 · BL PWM	

BOOT = GPIO0
RGB GPIO48 unused

CAN transceiver

TJA1051T/3 · VIO 3.3 V

CANH	
CANL	
CTX · D / TXD	
CRX · R / RXD	
VIO	3V3_P
VCC	5V
S · Silent	3V3_P
GND	GND

CTX → GPIO9 · S → VIO (silent)

NV3007 · 2.79″

428 × 142, SPI

3V3_D	VDD
GND	GND
	SCL
	SDA
	CS
	DC
	RES
	BL

SCL / SDA mean SPI here
MISO unused

BLE

Android phone · RU / EN

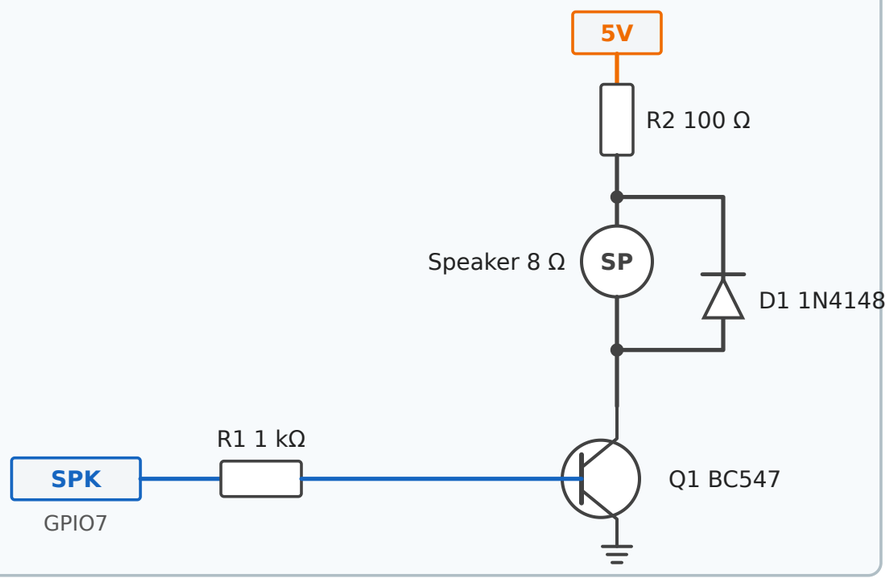
BLE: settings + OTA .bin
Bonded owner phone only

Wire colours

- 12 V
- 5 V
- Board 3V3_P
- 3V3_D DC-DC

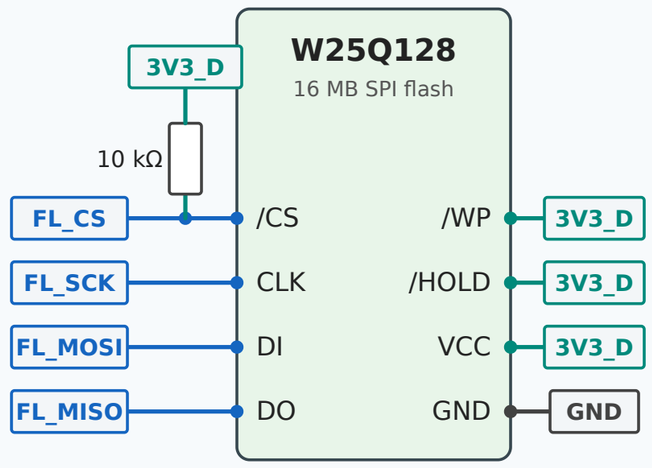
Speaker

Passive 8 Ω speaker through an NPN transistor



W25Q128 flash

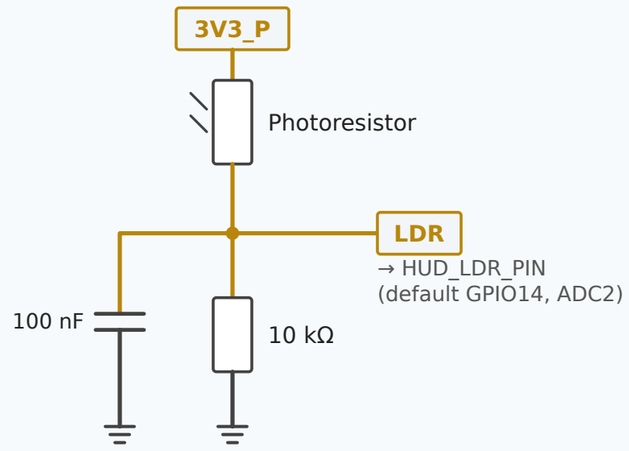
Camera database · powered from 3V3_D like the display



FL_CS = GPIO10 · FL_MOSI = GPIO11
FL_SCK = GPIO12 · FL_MISO = GPIO13

Light sensor (optional)

Used when the car gives no RLS_01 brightness



Pin is chosen by HUD_LDR_PIN in hud_config.h
ADC2 is fine: the firmware does not use Wi-Fi

Connections and notes

All GNDs common. LCD VDD → 3V3_D from 5 → 3.3 V DC-DC; input → 5V. CAN power from ESP32-S3: VCC → 5V; VIO → board 3V3.
BL / GPIO6 is a logic-level backlight control. If BL directly powers LEDs, use an external driver.
CAN: CRX → GPIO8, CTX → GPIO9 (TWAI TX). Receive-only: TWAI LISTEN_ONLY + S → VIO (silent).
Capacitors: 100 µF + 100 nF at board 3V3; 100 nF at CAN VCC; 47 µF + 100 nF at LCD VDD.
Use TJA1051T/3 with VIO: S → VIO (Silent). A version without /3 and with 5 V logic cannot be connected directly.
BOOT (GPIO0): hold for 5 seconds, then release. Pins are shown logically, not in their physical board positions.
Speaker: GPIO7 → 1 kΩ → BC547 base; speaker + 100 Ω from 5V to the collector; 1N4148 across the speaker (cathode → 5V).
W25Q128 on its own SPI bus: GPIO10 CS, GPIO11 MOSI, GPIO12 SCK, GPIO13 MISO. Power from 3V3_D like the display; CS pull-up 10 kΩ to 3V3_D; /WP and /HOLD → 3V3_D; 100 nF at VCC.
Light sensor (optional): photoresistor 3V3_P → pin, 10 kΩ pin → GND, 100 nF pin → GND. Pin set by HUD_LDR_PIN (default GPIO14, ADC2).